Onkar Jadhav

AI / Full-Stack Software Engineer
Pune, Maharashtra
B.Tech Computer Science, PCCOE (9.08 CGPA)

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Summary

Final-year CS student (9.08 CGPA) with hands-on project experience across production AI systems, machine learning, and full-stack engineering. Built Lore Engine, a production RAG platform with hybrid BM25 + pgvector retrieval over 6,000+ wiki pages and a deploy-gated evaluation harness. Designed StockXpert, a hierarchical deep learning prediction system achieving a 12% improvement in short-horizon return correlation across all 100 Nifty NSE equities. Built Stride, an AI-first adaptive wellness platform with five domain agents and multi-modal input. Proficient in Python, TypeScript, React, Next.js, FastAPI, Node.js, TensorFlow, RAG architectures, and agentic AI systems.

Education

•Bachelor of Technology in Computer Science *Nov 2022 – Jun 2026*
Pimpri Chinchwad College of Engineering, Savitribai Phule Pune University CGPA: 9.08 (up to 7th semester)

– Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

Engineering Projects

•Stride — *AI-First Adaptive Wellness Platform* *April 2026 – Present*
Next.js, Convex, OpenRouter, Multi-modal Input Processing, NLU

– Architected and shipped Stride to production, running live with an early private user cohort actively driving iterative product feedback ahead of public beta.
– Architected a six-layer AI-first system: multi-modal input capture, a behavioral memory layer, an NLU layer for structured extraction, five specialised domain agents (diet, workout, sleep, hydration, habits) sharing memory, an analytics layer, and an adaptive UX layer driven by behavioral signals.
– Separated concerns by design: LLMs handle understanding and intent extraction while a deterministic four-component calculator (BMR, NEAT, EAT, TEF) handles computation, eliminating opaque LLM arithmetic errors while retaining natural-language input.

•Lore Engine — *RAG Platform for Fan Wiki Knowledge Bases* *May 2026 – Present*
Next.js, FastAPI, PostgreSQL, pgvector, BM25, Celery, Redis, Cloudflare R2, OpenAI, Anthropic Claude

– Architected a full-stack RAG application ingesting **6,000+ MediaWiki pages** per series into a hybrid retrieval pipeline combining BM25 lexical search and pgvector dense embeddings via reciprocal rank fusion (RRF), chosen over pure vector search to handle proper-noun disambiguation and exact character-name matching that semantic models fail on.
– Designed an async ingestion pipeline (Celery workers, Redis broker, Cloudflare R2 object storage) that fetches pages from the MediaWiki API, converts wikitext to structured markdown, and persists infobox metadata in PostgreSQL.
– Established a **50-question per-series ground-truth eval harness** measuring retrieval MRR and end-to-end answer accuracy, gating every deploy; validated architecture against the Harry Potter wiki (∼6,000 pages).

•StockXpert — *AI-Powered Deep Learning Equity Prediction Platform* *Jan 2026 – May 2026*
Python, TensorFlow, FastAPI, Next.js, TypeScript, Tailwind CSS, Cloudflare R2, AWS

– Architected a hierarchical deep learning system (ResNet, GRU, Bi-LSTM, multi-head attention) modeling cross-horizon price dependencies across all **100 Nifty NSE equities**, achieving a **12% improvement in short-horizon return correlation**.
– Constructed a leakage-free feature pipeline from raw OHLCV data incorporating RSI, MACD, ADX, Bollinger Bands, and volatility indicators, with strict temporal train/validation/test splits to prevent lookahead bias.
– Built a snapshot-first serving layer on Cloudflare R2 with market-aware TTL caching, delivering sub-second API latency across all 100 Nifty stocks during market hours.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Objective-C, Java
Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS, Electron
Backend: FastAPI, Node.js, Express.js, BullMQ, Celery, Redis, REST APIs, WebSockets
AI / ML: TensorFlow, RAG Pipelines, pgvector, BM25, Reciprocal Rank Fusion, LLM Evaluation, Agentic Workflows, Behavioral Memory, Multi-modal AI, XGBoost, OpenRouter
LLM Providers: Anthropic Claude, OpenAI GPT, Gemini, Groq
Databases: PostgreSQL (+ pgvector), MongoDB, Convex
Cloud & DevOps: AWS, Cloudflare R2, Git, GitHub
System Design: Event-driven architecture, async job pipelines, hybrid retrieval, multi-agent coordination